Ruhan Wang

Computer Engineering Ph.D. | LLM Agents, Reinforcement Learning & Scalable Post-Training

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PROFESSIONAL PROFILE

Computer Engineering Ph.D. working on **long-horizon LLM agents, self-evolving agent harnesses, reinforcement learning for LLMs, and post-training**. Experienced in RLVR, policy optimization, and improving RL training efficiency. Builds self-evolving agents that use tools, interact with environments, complete long-horizon tasks, and improve through execution feedback. Research has been published at leading conferences and journals, including ICML, ICLR, COLM, and TMLR.

EDUCATION

Indiana University — Aug. 2022 to Feb. 2027 (expected)
Ph.D. in Computer Engineering — *Bloomington, IN*

Indiana University — Aug. 2022 to Dec. 2024
M.S. in Computer Engineering — *Bloomington, IN*

CORE EXPERTISE

Agentic RL, Evals & Environments	Tool-using terminal agents; long-horizon execution; agent environments and runtime harnesses; evals, dense-reward graders, and reward signals; multi-agent coordination; failure and model-behavior analysis.
LLM Post-Training & RL	RLVR and policy optimization with GRPO and W-REINFORCE; verifiable rewards; dynamic batch scheduling; reasoning optimization; training-signal and model-behavior analysis.
Scalable Training & Systems	Python, PyTorch, Hugging Face Transformers, veRL, vLLM, Ray, and DeepSpeed; distributed multi-GPU training and inference; Docker, Apptainer, Slurm, and GPU clusters.

EXPERIENCE

Tencent AI Lab | Hunyuan Frontier Lab — May 2026 to Present
Ph.D. Research Intern | Agentic Reinforcement Learning; Self-Evolving Agents — *Bellevue, WA*

- **Long-horizon agent harness interpretability:** Introduced the Harness Handbook, a behavior-centric representation that maps high-level system behaviors to concrete implementation sites, making complex agent harnesses easier to navigate, audit, and modify.
- **Self-evolving long-horizon agent harnesses:** Built an execution-feedback loop that identifies failures, proposes harness changes, and validates their effects, enabling iterative improvement from observed agent behavior.
- **Long-horizon agent evaluation:** Co-developed Long-Horizon-Terminal-Bench, a benchmark of 46 tasks across nine categories, and evaluated 15 frontier models on runs averaging 9.9M tokens, 231 episodes, and 85.3 minutes.
- **Long-horizon agent task synthesis:** Co-developed Recursive Synthetic Terminal Tasks, a framework that recursively generates and validates long-horizon tasks in fresh sandboxes, producing 37,484 verified tasks across 15 rounds at approximately $0.05 per task. Agentic PPO on the synthesized data improved Qwen3.5-27B by 20.0%, 41.2%, and 21.9% on Terminal-Bench 2, Terminal-Bench Hard, and Long-Horizon Terminal Bench.
- **Long-horizon terminal-agent RL:** Co-developed T1, a 122B Mixture-of-Experts agent post-trained with reinforcement learning for real-shell tasks of more than 300 tool-call turns. Dense process rewards and TITO/R3 training-inference alignment raised the Terminal-Bench 2.1 resolved rate from 43.8% to 64.0% and surpassed Gemini 3.1 Pro and GLM 5.1 on Long-Horizon Terminal Bench.

Indiana University | Machine Learning Lab — Aug. 2023 to Present
Graduate Researcher; Advisor: Prof. Dongruo Zhou — *Bloomington, IN*

- **Efficient RLVR:** Identified sparse gradient signals in LLM reinforcement learning and developed dynamic batch scheduling, reducing optimizer updates by up to 79% and wall-clock time by 9.3% while preserving Pass@k across three LLMs, four benchmarks, and three RL algorithms.
- **Federated LLM reasoning:** Developed FERA to combine reasoning from distributed LLMs without training, outperforming baselines on MMLU-Pro, AQUA-RAT, and GSM8K while using approximately 88% fewer FLOPs than FedAvg

and 64% fewer than FLoRA.

- **Collaborative LLM agents:** Developed Fed-ICL to improve answers without sharing model parameters or raw data, increasing MMLU accuracy by up to 11% and TruthfulQA BERTScore by up to 8% while converging in 40% to 88% fewer rounds than FedAvg.
- **Multimodal agentic recommendation:** Designed an architecture that integrates user profiles, memory, planning, actions, multi-agent collaboration, and interactive user simulation, and identified seven open challenges for next-generation systems.
- **Offline and safe RL:** Developed RADT for policy transfer across changing dynamics and CQDT for trajectory stitching under safety constraints, with theoretical guarantees and evaluations on D4RL and safe offline RL benchmarks.

Mitsubishi Electric Research Laboratories — May 2024 to Aug. 2024

Ph.D. Research Intern | Generative AI — *Cambridge, MA*

- **Generative AI for few-shot learning:** Developed three label-guided quantum diffusion algorithms that achieved 79.5% mean accuracy, a 17.3% relative improvement over the strongest QNN baseline across 12 tasks; validated them on noisy IBM Almaden hardware, leading to publications at ICAD 2025 and the AAAI QCAI Workshop.

SELECTED PUBLICATIONS

Selected papers are listed below; see Google Scholar for the complete publication record.

Agentic AI & LLM Post-Training

[1] Junyao Yang, Yucheng Shi, **Ruhan Wang**, Zhongzhi Li, Zongxia Li, Haitao Mi, and Leowei Liang. "T1: Terminal Agent Reinforcement Learning for Long-Horizon Tasks." 2026.

[2] **Ruhan Wang**, Kishan Panaganti, and Dongruo Zhou. "More Memory, Worse Agents: Error Reproduction and Anti-Persistence in LLM Agents." *Under review at NeurIPS 2026*.

[3] Yue Yu, Runze Zhao, **Ruhan Wang**, David J. Crandall, Yinglun Zhu, and Dongruo Zhou. "Computationally Efficient Reinforcement Learning with Verifiable Rewards via Dynamic Batch Scheduling." *Under review at NeurIPS 2026*.

[4] **Ruhan Wang**, Yucheng Shi, Zongxia Li, Zhongzhi Li, Yue Yu, Junyao Yang, Kishan Panaganti, Haitao Mi, Dongruo Zhou, and Leowei Liang. "Harness Handbook: Making Evolving Agent Harnesses Readable, Navigable, and Editable." *arXiv:2607.13285*, 2026.

[5] Zongxia Li, Zhongzhi Li, Yucheng Shi, **Ruhan Wang**, Junyao Yang, Zhichao Liu, Xiyang Wu, Anhao Li, Yue Yu, Ninghao Liu, Lichao Sun, Haitao Mi, and Leowei Liang. "Long-Horizon-Terminal-Bench: Testing the Limits of Agents on Long-Horizon Terminal Tasks with Dense Reward-Based Grading." *arXiv:2607.08964*, 2026.

[6] Junyao Yang, Yucheng Shi, Zongxia Li, Zhongzhi Li, **Ruhan Wang**, Xiangxin Zhou, Kishan Panaganti, Haitao Mi, Leowei Liang. "Stale but Stable: Staleness-Adaptive Trust Regions for Stabilizing Asynchronous Reinforcement Learning Authors" *arXiv:2607.18722*, 2026.

[7] Zhongzhi Li, Yucheng Shi, Zongxia Li, **Ruhan Wang**, Anhao Li, Zixun Huang, Junyao Yang, Lei Ke, Ninghao Liu, Haitao Mi, Leowei Liang. "Recursive synthesis for long-horizon terminal tasks." *arXiv:2608.05466*, 2026.

Collaborative & Multimodal LLMs

[8] **Ruhan Wang**, Chengkai Huang, Zhiyong Wang, Junda Wu, Rui Wang, Tong Yu, Julian McAuley, Lina Yao, and Dongruo Zhou. "FERA: Uncertainty-Aware Federated Reasoning for Large Language Models." *COLM 2026*.

[9] **Ruhan Wang**, Zhiyong Wang, Chengkai Huang, Rui Wang, Tong Yu, Lina Yao, John C. S. Lui, and Dongruo Zhou. "Federated In-Context Learning: Iterative Refinement for Improved Answer Quality." *ICML 2025*.

[10] Chengkai Huang, Junda Wu, Yu Xia, Sheldon Yu, **Ruhan Wang**, Tong Yu, Ruiyi Zhang, Ryan Rossi, Branislav Kveton, Dongruo Zhou, Julian McAuley, and Lina Yao. "Towards Agentic Recommender Systems in the Era of Multimodal Large Language Models." *ACM TIST 2025*.

Reinforcement Learning & Efficient AI

[11] Runze Zhao, Yue Yu, **Ruhan Wang**, Chunfeng Huang, and Dongruo Zhou. "Instance-Dependent Continuous-Time Reinforcement Learning via Maximum Likelihood Estimation." *ICML 2026*.

[12] Zhishuai Liu, Yu Yang, **Ruhan Wang**, Pan Xu, and Dongruo Zhou. "How to Provably Improve Return Conditioned Supervised Learning?" *Under review at NeurIPS 2026*.

[13] **Ruhan Wang**, Yu Yang, Zhishuai Liu, Dongruo Zhou, and Pan Xu. "Return Augmented Decision Transformer for Off-Dynamics Reinforcement Learning." *TMLR 2025*.

[14] **Ruhan Wang** and Dongruo Zhou. "Safe Decision Transformer with Learning-Based Constraints." *L4DC 2025; NeurIPS*

Safe Generative AI Workshop 2024.

[15] Ahmad Faiz, Sotaro Kaneda, **Ruhan Wang**, Rita Osi, Parteek Sharma, Fan Chen, and Lei Jiang. "LLMCarbon: Modeling the End-to-End Carbon Footprint of Large Language Models." *ICLR 2024 (Oral)*.

Quantum Computing

[16] **Ruhan Wang**, Philip Richerme, Fan Chen. "A hybrid quantum–classical neural network for learning transferable visual representation." *Quantum Science and Technology, 2023*.

[17] **Ruhan Wang**, Fahiz Baba-Yara, Fan Chen. "Justq: Automated deployment of fair and accurate quantum neural networks Authors." *ASP-DAC 2024*.

[18] **Ruhan Wang**, Ye Wang, Jing Liu, Toshiaki Koike-Akino. "Quantum diffusion models for few-shot learning." *ICAD 2025*.

PROFESSIONAL RECOGNITION & SERVICE

- **Conference reviewing:** ICML (2025, 2026; **Gold Reviewer**, 2026); NeurIPS (2025, 2026);
  AAAI (2025, 2026, 2027); ICLR (2026, 2027; **Top 25% Reviewer**, 2026).
- **Journal reviewing:** *Measurement Science and Technology*;
  Transactions on Machine Learning Research (TMLR).
- **Teaching:** Teaching Assistant / Associate Instructor
  Reinforcement Learning for LLMs (Spring 2026); Principles of Machine Learning (Spring 2025);
  Data Mining (Fall 2024, Fall 2025); Engineering Cloud Computing (Spring 2024);
  Deep Learning Architecture and Hardware Acceleration (Fall 2023).